

Fig. 3: Visualising pole and throw, NO-NC concepts

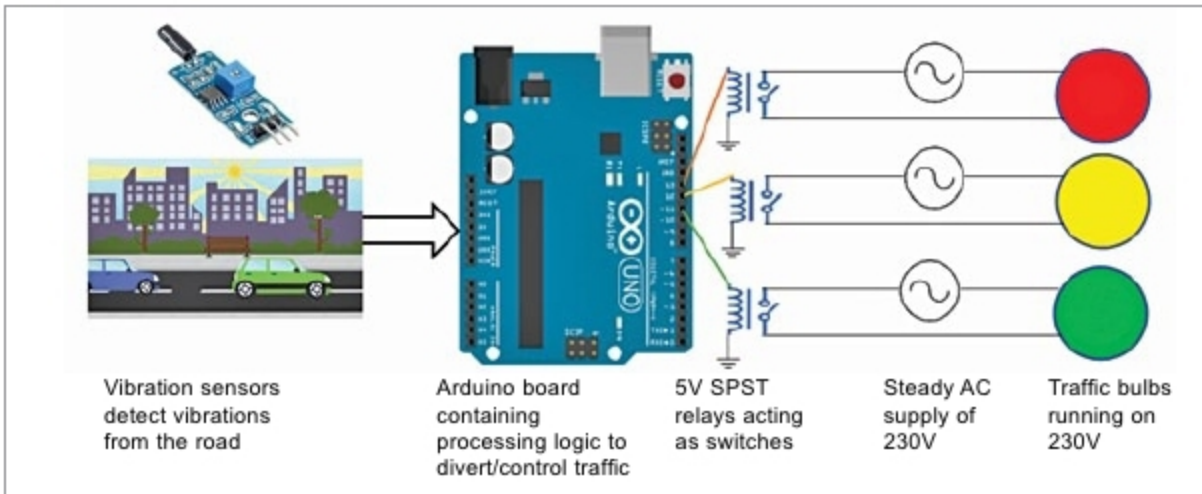


Fig. 4: Controlling traffic signals

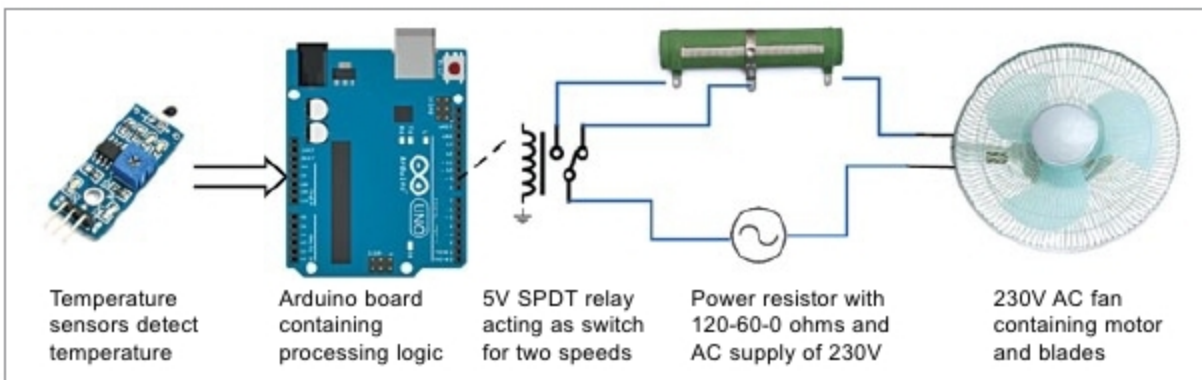


Fig. 5: Temperature-based control of table fan

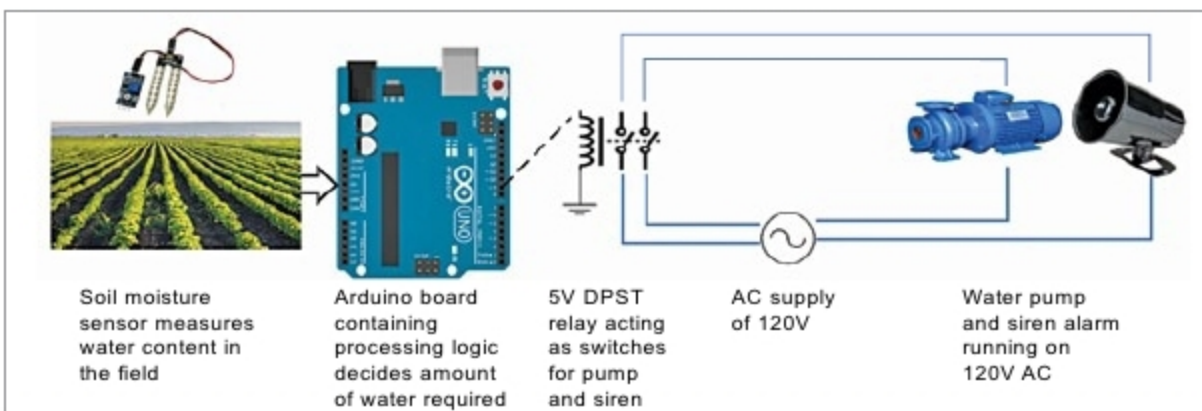


Fig. 6: Precision agriculture

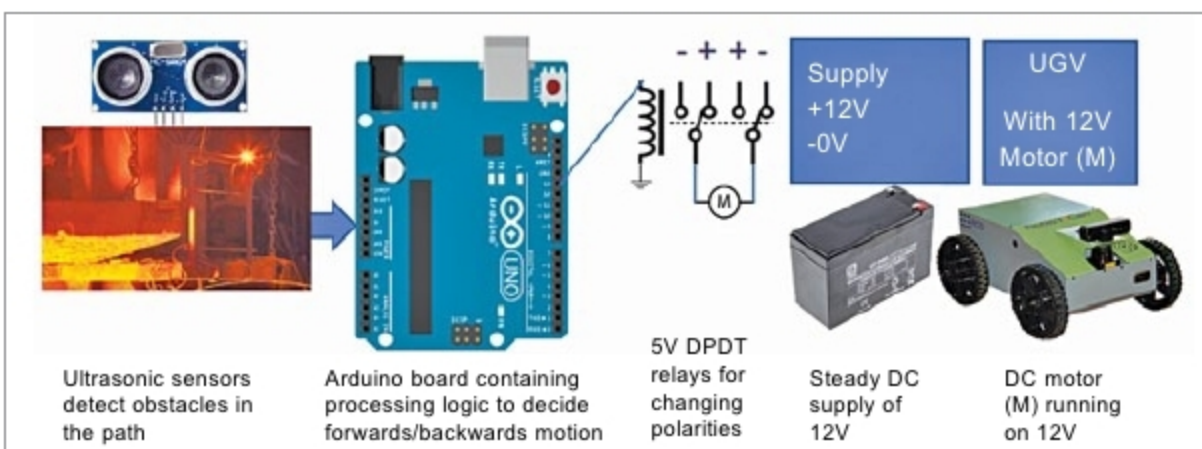


Fig. 7: UGV for inhospitable environments

Application 2: Temperature-based control of table fan (using SPDT relay).

Application 2 shows how to control the speed of a table fan working on 230V AC supply. Speed of DC motors or fans can be controlled by using simple potentiometer or pulse-width modulated (PWM) signals. For AC equivalents, controlling can be done using power resistors or power semiconductor devices.

A power resistor can withstand the rated wattage on it. Resistance across shown power resistor varies from 120 ohms to 60 ohms to 0 ohms, left to right with the right-most terminal fixed. Depending on temperature, an SPDT relay can be switched on to provide resistance of 120 ohms or 60 ohms in the circuit, leading to different fan speeds.

Application 3: Precision agriculture (using DPST relay). Application 3 shows how to water a field according to the water content in the soil. Arduino decides the amount of time and, hence, the amount of water to be pumped in the field, using a DPST relay. A siren in parallel sounds to indicate pump is on condition. This application is helpful in saving water.

Application 4: Unmanned ground vehicle for inhospitable environments (using DPDT relay). Application 4 explains how to control a simple unmanned ground vehicle (UGV) that moves forwards and backwards depending on input. A DC motor's direction of rotation is controlled by alternating the polarities of input voltage. A sensor such as ultrasonic sensor observes the obstacles and decides whether to move forwards or backwards. Fig. 7 shows a DPDT relay having two switches controlled by one excitation signal. Switches with polarity of 12V are required for the motor. Direction of motion of the UGV changes accordingly.

These applications indicate viable methods to employ relays to generate real-world applications. Relays, either electromagnetic, reed or solid-state, perform the same operation, that is, switching. Electromagnetic relays are easily implementable for any voltages or any devices and, hence, should help people in demonstrating projects successfully for environmental, medical, military, agriculture, societal, automobile, aviation and other applications. **EFY**